

Smart SWIPE: Active Learning-based Intelligent Photo Organization System

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Author: Siddharth Kumar (Roll No. 240403)

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Abstract

Modern smartphones store thousands of images, making manual organization difficult. Smart SWIPE is an Active Learning-based image organization system that learns user preferences from around 300 swipe interactions and automatically classifies more than 5,000 images. The system minimizes manual labeling while continuously improving through user feedback.

1. Motivation

Traditional gallery applications require manual sorting and album creation. Smart SWIPE leverages Active Learning and deep learning to reduce user effort while maintaining high classification accuracy.

2. Problem Statement

Develop an intelligent image classification system capable of learning user preferences from a small number of swipe interactions and organizing large photo collections automatically.

3. Objectives

- Learn user preferences using ~300 swipes.
- Automatically organize 5,000+ photos.
- Reduce manual labeling effort.
- Continuously improve through user feedback.

4. Methodology

Pipeline: Image Collection → Preprocessing → Feature Extraction (CNN) → Active Learning → User Swipe Feedback → Model Training → Automatic Photo Organization.

5. Technologies Used

Python, PyTorch, OpenCV, NumPy, TorchVision, Scikit-learn, Matplotlib, Jupyter Notebook.

6. Evaluation Metrics

Accuracy, Precision, Recall, F1-score, Number of Swipes, Label Reduction, Classification Confidence, Inference Time.

7. Expected Results

The proposed system aims to classify over 5,000 images after only ~300 user interactions, significantly reducing manual effort while improving classification quality.

8. Applications

Smartphone gallery organization, cloud photo management, automated album creation, digital assistants, multimedia retrieval.

9. Future Scope

Face recognition, event detection, duplicate removal, multimodal image understanding, natural language image search.

10. Conclusion

Smart SWIPE demonstrates how Active Learning can significantly reduce manual effort in photo organization by learning from minimal user feedback and scaling to large image collections.